Allaye GUINDO

Geomatics Specialist · GIS & Remote Sensing

Geomarketing · Environment · Water Resources · Energy

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Portfolio & interactive maps: **guindoallayecontact7.github.io**

PROFILE

Geomatics and GIS specialist with over 10 years of experience in spatial analysis, remote sensing, field data collection and monitoring & evaluation, including over 4 years as Geomarketing Manager at WeLight (AXIAN Group), a rural electrification operator deploying solar mini-grids in more than 8 African countries. PhD candidate in remote sensing applied to natural resource management (Sahelian wetlands, Lake Wégnia). Contributor to environmental and social impact studies (ESIA / ESMP) and flood risk assessments. Experienced in coordinating multidisciplinary, multi-country teams and producing decision-support analyses and reports.

KEY SKILLS

GIS & remote sensing	ArcGIS Pro, QGIS, GRASS GIS, ENVI, OpenStreetMap, Sentinel-2 imagery
Environment & water	ESIA / ESMP studies, wetlands, soil erosion (RUSLE), water quality, flood risk
Geomarketing	Multi-criteria analysis of electrification potential, demand and socio-economic studies
Data & analytics	Python, R, Power BI, SPSS, XLSTAT, advanced Excel (macros), Access
Digital data collection	KoboToolbox / KoboCollect, ODK, Ona, CSPro, SurveyCTO
Quality & M&E	Automated quality control, standard operating procedures, data governance, KPIs, reporting
Management	Project management, multi-country field team coordination, training
Languages	French and English: bilingual · Bambara, Dogosso, Fulfulde

PROFESSIONAL EXPERIENCE

Geomarketing Manager — WeLight (AXIAN Group)
April 2022 – present

Bamako, Mali · Mali, Nigeria, Madagascar, DRC, Malawi, Uganda, Republic of the Congo, Cameroon…

- Coordinate geomarketing studies and geospatial analyses of rural electrification potential in more than 8 African countries (expanding portfolio).
- Contribute to the environmental and social impact studies (ESIA) of 14 rural electrification sites in Mali, with the E&S teams.
- Set up automated quality-control procedures for databases and a standardised multi-country coding protocol.
- Supervise field surveys and energy demand studies (Mali, Nigeria, DRC); train local teams on GIS tools.
- Design Power BI dashboards to monitor data quality and the performance of potential sites.

GIS Consultant — Save the Children
2022

Bamako, Mali

- Trained safety, security and MEAL staff in GIS: geocoding, geoprocessing, thematic mapping.
- Designed geospatial data collection and monitoring tools for humanitarian settings; integrated GIS into MEAL activities.

Monitoring & Evaluation Consultant — FAWE Mali & Zanzibar
February – March 2022

- Socio-economic evaluation of the ETP project: data collection tools (KoboCollect), analysis, recommendations and final report.

Data Manager Consultant — ARCAD SIDA
March – December 2021

Bamako, Mali

- Set up a digital data collection system (KoboToolbox) and trained field teams.
- Data cleaning, quality control and analysis; decision-support reporting.

Mapping Team Leader — INSTAT – National Institute of Statistics
February 2019 – February 2021

Koulikoro and Ségou regions

- General Census of Economic Units (RGUE): supervised mapping, listing and enumeration.
- Data quality control and validation; field team supervision and community relations.

Research Assistant · Project Lead — Labo-HoPE, FHG – USSGB
2015 – 2019

Bamako, Mali

- Project lead (Oct.–Dec. 2019): multi-source flood risk study (remote sensing, fieldwork) and prevention recommendations.
- Studied past and current floods (Bamako, Djenné, Bla) and Niger River dynamics; contributed to publications.

Research Assistant (freelance) — Research West Africa
October 2015 – April 2019

Bamako, Mali

- Market research surveys for an international firm: data collection, field supervision and data quality monitoring.

Other assignments — Mapping & surveys
2017 – 2019

- **Ministry of Culture / DNPC, 2019:** inventory and mapping of archaeological sites in Mali, funded by the U.S. Embassy.

- **Sagaci Research, 2019:** Ona/ODK questionnaires, dashboards, supervision of B2B/B2C surveys and mystery shopping (Mopti).
- **SERTAS, 2017 – 2018:** mapping of stormwater drainage networks for the PGRCI project (Bamako, Kayes, Mopti).

EDUCATION

PhD in Geomatics — remote sensing applied to natural resource management *Since April 2025*
USSGB, ED-DESSLA Doctoral School · *Thesis: spatio-temporal dynamics of water and environmental resources in Sahelian wetlands, the case of Lake Wégnia (Mali)*

Master's degree (M2) — Geomatics, Spatial Planning and Territorial Management (Géo-GATE) *2018 – 2020*
USSGB, Faculty of History and Geography · *with honours (Mention Bien)*

Bachelor's degree — Physical Geography and Environment *2014 – 2017*
USSGB, Faculty of History and Geography

ADDITIONAL TRAINING & CERTIFICATIONS

- **GMES and Africa (2024):** sustainable management of wetlands and floods for food security and ecosystem resilience in West Africa
- **Digital Earth Africa (2025 – 2026):** wetlands workflows · water resources · urbanisation · land degradation · IWMI water data products · platforms and Sandbox
- **Online certifications:** Geospatial Raster Data Analytics in Python · Python: Predictive Analytics · Power BI (Power Apps & Power Automate) · Data Science Foundations · PostgreSQL Essentials · Managing an International Team
- **Upgrade Africa (2023):** Microsoft Office & professional writing
- **Bamako Development Center (2018):** Personal leadership
- **Labo-HoPE, USSGB (2016):** CAD, mapping and GIS (Inkscape, QGIS); statistical methods and scientific writing

PUBLICATIONS & SCIENTIFIC PRESENTATIONS (IN FRENCH)

- Étude de la contribution des forages dans l'amélioration de l'accès à l'eau potable à N'Tabacoro Cité Extension (Bamako) [Contribution of boreholes to drinking water access]. European Scientific Journal, 2021. doi.org/10.19044/esj.2021.v17n40p106
- Guindo A., Diawara H., Berthé T. — Évaluation spatio-temporelle de la turbidité des eaux du lac Wégnia à partir de l'imagerie satellitaire Sentinel-2 (Mali) [Spatio-temporal assessment of Lake Wégnia turbidity using Sentinel-2]. 15th Malian Symposium on Applied Sciences (MSAS 2026), Bamako, oral presentation.
- Analyse des paramètres morphométriques et du risque d'inondation dans les bassins versants de Niamakoro et Missabougou [Morphometric parameters and flood risk], 2022. urls.fr/eRpx-e
- Berthé T., Diawara H., Bengaly S., Guindo A. — Cartographie des zones à risque d'inondation dans le bassin versant de la rivière Woyowayanko à Bamako [Flood risk mapping, Woyowayanko watershed]. MSAS 2026, Bamako.
- Coulibaly K., Guindo A. — Célébration ostentatoire des cérémonies sociales en milieu urbain : cas du mariage dans le district de Bamako [Ostentatious wedding ceremonies in Bamako], 2022. urls.fr/ILM11v

Posters at the International GIS Days (JISIG, FHG – USSGB):

- 2020 — Remote sensing and GIS for flood risk assessment: the Missabougou watershed.
- 2020 — Central Mali crisis and access to healthcare: the Bankass district.
- 2019 — Channel mobility of the Bani River between Sofara and Tié.
- 2019 — Mapping illegal waste dumping sites in riverbeds of the Bamako district.

VOLUNTEERING & AWARDS

- **Humanitarian mapping — HOT:** Task Manager · Advanced Mapper (since 2017); YouthMappers FHG, trainer (2018 – 2022).
- **Faculty of History and Geography:** co-organiser of the International GIS Days (JISIG), 4th (2019) and 5th (2020) editions.
- **Labo-HoPE:** certificate of recognition for contributions to the laboratory's scientific development (2020).

REFERENCES

References available upon request.